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Top Skills

jupyter
NumPy
Pandas

Certifications

ECPC - Egyptian Collegiate
Programming Contest 2025
Python Programmer Bootcamp
Sprints x Microsoft Summer Camp -
AI and Machine Learning
Introduction to Python

Honors-Awards

1st rank in Statistics and Computer
Science program (mathematics
Department)
1st rank in Statistics and Computer
Science program (mathematics
Department)

Sofyan Kirat

Data Science Trainee | Problem Solver & Competitive Programmer
| Graphic & Logo Designer | ALX Alumni – Software Engineer &
Backend Specialist

El Mansoura, Ad Daqahliyah, Egypt

Summary

I am Sofyan Kirat, an aspiring Data Scientist with a strong foundation in mathematics, statistics, and computer science. I aim to use this expertise to turn complex data into actionable insights, enabling data-driven decisions and creating lasting impact.

Educational Background & Achievements:

I am a student at Mansoura University, Faculty of Science, majoring in Statistics and Computer Science. For three consecutive years, I ranked 1st in my department and graduated the past two years with Excellent with honors. My studies in mathematics, statistics, and computer science have sharpened my analytical thinking, strengthened my logic, and equipped me with strong technical skills that drive my career forward.

Professional Experience & Internships:

Before pursuing data science, I completed the ALX Software Engineering Program – Backend Specialist, a 14-month intensive training that strengthened my software development and problem-solving skills. For my graduation project, I joined the Codragons team to build DocMate and DocVice - health platforms providing medical guidance and verified education. The work involved managing datasets, analyzing information, and creating data-driven features, which revealed the transformative power of data in solving real-world challenges and inspired my transition into data science, merging my foundation in mathematics and statistics with advanced technologies.

Passions, Interests & Skills:

Beyond data science, I bring expertise in software engineering, backend development, and full-stack project execution, along with over three years in graphic design and logo creation. I enjoy competitive programming, problem-solving, and actively joining

related competitions. Driven by a passion for helping others and sharing knowledge, I believe in lifelong learning and inspiring others to grow their skills and curiosity.

Vision & Future Goals:

My long-term goal is to become both a leading Data Scientist and a professor at my university, teaching mathematics, statistics, and computer science in a way that ensures deep understanding. I aim to contribute to AI, machine learning, and statistical modeling innovations while making these sciences more accessible through education. Currently, I am dedicating myself to mastering all core areas of data science, AI, and machine learning while excelling in my academic journey.

If you are interested in collaborating, seeking guidance, or simply connecting to exchange ideas, I would be happy to hear from you. Feel free to reach me at www.sofyankirat@gmail.com.

Experience

Sprints

AI & ML intern

July 2025 - September 2025 (3 months)

During this intensive Summer Camp, I developed strong practical and theoretical skills in Artificial Intelligence and Machine Learning, culminating in the successful delivery of a complete end-to-end ML graduation project.

Throughout the program, my work covered:

- Python fundamentals, data structures, and object-oriented programming
- Core Python data-science libraries such as NumPy, Pandas, Matplotlib, Seaborn, and Plotly
- Key supervised learning methods including Linear Regression, Random Forest, and Support Vector Machines
- Foundational unsupervised learning techniques including K-means and DBSCAN, along with PCA and cross-validation strategies

Graduation Project – Heart Disease Prediction:

Using a dataset of 14 clinical features and 303 patient records, the project focused on predicting the presence of heart disease through a structured ML workflow that involved:

- Exploratory Data Analysis supported by meaningful visualizations
 - Comprehensive data preprocessing including handling missing values, feature scaling, and categorical encoding
 - Feature selection and dimensionality reduction using PCA, RFE, Chi-Square tests, and model-based feature importance
 - Model development using Logistic Regression, Decision Tree, Random Forest, and SVM classifiers
 - Achieving a final accuracy of 93.4% following systematic hyperparameter tuning
 - Applying K-means and Hierarchical Clustering to uncover hidden structure within the data
 - Evaluating model performance using precision, recall, and F1-score
- Project Repository: <https://github.com/sofyankirat/Comprehensive-Machine-Learning-Full-Pipeline-on-Heart-Disease-UCI-Dataset>

alx_africa

Software Engineer

November 2023 - February 2025 (1 year 4 months)

Nairobi County, Kenya

ALX is a mission-driven tech education initiative run by the ALX Foundation to empower African learners with world-class, future-ready skills, combining technical expertise with professional competencies like critical thinking, teamwork, and communication to ensure job readiness.

Graduated from the ALX Software Engineering Program – Backend Specialist, a 14-month immersive, project-based training preparing aspiring engineers for the tech job market through real-world problem-solving, self-directed learning, and collaborative teamwork. The curriculum progressed from foundational programming to advanced backend specialization, covering industry-standard tools and technologies.

1- Foundations Phase (Nov 2023 – Sep 2024):

- >C language and data structures using C
- >Python language and OOP using python

- >JavaScript language
- >Database using SQL
- >DevOps basics.
- >Built projects including an Airbnb clone.
- >Applied Git/GitHub for version control and collaboration.

2- Backend Specialization (Sep 2024 – Feb 2025):

- >Node.js backend development.
- >Advanced MySQL & NoSQL (MongoDB) databases.
- >Queueing systems
- >file management
- >secure user data handling.

3- Extended Learning – Frontend Development (Reward Track)

Gained exclusive access to the Frontend Specialization as a graduation reward.

- >HTML
- >CSS and responsive design
- >React to create modern, interactive, and user-friendly interfaces.

For my graduation project, I collaborated in a 6-member team (4 Frontend Specialists and 2 Backend Specialists, including myself) called “Codragons” to develop DocMate and DocVice — intelligent health platforms designed to predict early disease risks, guide timely medical consultations, and provide verified health education to combat misinformation and improve healthcare accessibility.

My ALX journey strengthened my full-stack expertise and honed my problem-solving, collaboration, and project management skills, equipping me to deliver impactful, real-world software solutions.

Education

Mansoura University

Bachelor of Science - BS, Statistics and Computer Science · (October 2022 - June 2026)

